Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Xavier Tan	tanx@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

Because I am chemicalEngineering I will attempt to have something that takes balanced equations and starting reactant masses and then can calculate limiting reactant, theoretical yield and percent yield based on said principles from above.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

There is already sites such as Khan academy and other chemistry sites that already do this

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

To generally check my work in chemistry so it will have practical application and must work because I do not want false answers

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

Being able to intake equations via floats and strings and then understand them to output values or ratios to use for specific problems user can input as well

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

Equation balanced, either moles or grams(can be question)

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

It'll use the equation it has and understand the coefficients to then calculate grams or moles of reactants and products using and comparison results

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

Depending on what the user gives and wants we will make it so it gives what they want

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Chemical, reactants, products

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Chemical formula/name, molar mass, stoich coefficient, and mass/moles. Most other things such as state I will ignore despite being important normally or charges

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

Most likely due to storing and accessing it can be easy to get lose the actual formulas are well known thus shouldn't be too hard.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

Ill most likely be breaking it down into helper functions as well as I will personally outline how I want it to be organized that way if I do use an LLN its mainly meant to just fill in the holes ive made

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

To try to use as also proof i understand both code and chemistry stoich

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to build a simple web tool to help students do their homework as well as check their work and I will try to paste the formulas so they understand what the code did as well. What are 3 common computational challenges students face when doing stoich, and what are 3 core features a basic MVP tool should have?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

Key challenges include stringing of chemical formulas, floating-point precision errors during mole calculations, and handling reactions with multiple products. It recommended keeping the MVP focused on three things: a clean input form for entering balanced equations and masses, a robust function to compute moles and identify the limiting reagent, and a clear output view displaying theoretical and percent yields.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

Using advanced regex libraries or external APIs to balance chemical equations and trying to make it capable of doing mass equations or several equations is beyond a simple aspect. What is a simpler alternative where the user inputs coefficients or predefined compounds?

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

I know it must be simple relatively towards chemistry because it gets complicated fast

Part 6: Resource Planning

1. What resources will you use if you get stuck?

LLM, friends or TAs

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

Debugging myself or showing the LLM something went wrong

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Conversation with Gemini

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Role

Name

Email

Developer (You)

Xavier Tan

tanx@udel.edu

Friend (optional)

Part 1: The Goal & Prior Implementations

1. Identify the Goal: What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.

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- Otherwise, create a website related to your BMEG100 Lab Report 1, Cytoskeletal and Nuclear Staining Lab Report Guidelines, linked here (University of Delaware account required to access).
- Chemistry and Chemical Engineering majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
- Majors in the Electric & Computer Engineering department may consider a steganography project that hides messages in images, and then retrieves

those messages from uploaded images.
- Mechanical Engineering majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

Because I am chemicalEngineering I will attempt to have something that takes balanced equations and starting reactant masses and then can calculate limiting reactant, theoretical yield and percent yield based on said principles from above.

1. Prior Implementations: Search the internet. What existing tools or websites currently attempt to address it?

There is already sites such as Khan academy and other chemistry sites that already do this

1. Your Implementation: Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

To generally check my work in chemistry so it will have practical application and must work because I do not want false answers

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. The Core Goal: Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

Replace this text with your answer.

1. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

Replace this text with your answer.

1. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

Replace this text with your answer.

1. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

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Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. Representing Reality: Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., If building a recipe app, the nouns

are "Recipe" and "Ingredient").

Replace this text with your answer.

1. Identifying Properties: For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?

- Details I need: (e.g., Ingredient name, weight, cost)
- Details I will ignore: (e.g., Ingredient color, brand, expiration date)

Replace this text with your answer.

Part 4: Challenges, AI Strategy, & Goals

1. Difficulties: What parts of this project seem the most difficult or technically complex?

Replace this text with your answer.

1. Coding Strategy: If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

Replace this text with your answer.

1. Goals: What is your ultimate goal for this project? (e.g., Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply).

Replace this text with your answer.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. The Initial Prompt/Question: Ask the AI or your friend to help you brainstorm. (e.g., "I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"). Paste your prompt below:

Replace this text with your answer.

1. The Response: Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

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1. The Pushback: You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain

why it is too complicated for an introductory programming project. Ask them for a simpler alternative. Paste your response here:

Replace this text with your answer.

1. Reflection: Based on this conversation, how has your project idea changed or become more focused?

Replace this text with your answer.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

Replace this text with your answer.

1. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

Replace this text with your answer.

Part 7: LLM Conversation

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Replace this text with your answer.

Part 8: Submit

In Google docs, with this document open, please click on File > Download > Microsoft Word (.docx), and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.

Computer Science - Week 10: Final Project - Pick a Project

Role: Developer (You)
Name: Xavier Tan
Email: tanx@udel.edu

Part 1: The Goal & Prior Implementations

Identify the Goal: What specific goal are you trying to address?
Because I am a Chemical Engineering major, I will build an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and harvested product amounts to calculate the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e *(OpenStax) Chapter 4. Additionally, it will compute reaction enthalpy to determine if a reaction is exothermic or endothermic and calculate the heat flow per mole of product.*
Prior Implementations: Search the internet. What existing tools or websites currently address it?
There are existing educational sites such as Khan Academy and various online stoichiometry calculators that handle

basic mole conversions and balancing, but they rarely combine step-by-step limiting reagent determination, percentage yield tracking, and thermodynamic heat flow calculations into a single interactive web application.

Your Implementation: Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

I want this tool to serve as a reliable calculator to check my work in chemistry coursework and lab calculations, ensuring practical accuracy and preventing false answers by executing rigorous stoichiometric logic.

Part 2: The Input-Process-Output (IPO) Model

The Core Goal: In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The program takes user-defined reactant masses and balanced chemical equations as input strings and floats, processes them through stoichiometric mole-ratio comparisons and thermodynamic formulas, and outputs an organized breakdown of limiting reagents, product yields, and reaction enthalpy.

Input: What specific information will the user need to provide to your program?

The user will provide a balanced chemical equation (or individual reactants and products), the starting mass (in grams) for each reactant, and the actual harvested mass of the product.

Processing: At a high level, how will the computer manipulate or evaluate that data?

The computer will parse molecular formulas into molar masses, convert given reactant masses into moles, evaluate mole-to-coefficient ratios to identify the limiting reactant, compute theoretical yields, compare actual yield to theoretical yield to find the percent yield, and multiply moles of product by the heat of reaction to compute total heat flow.

Output: What will the user get back?

An organized breakdown showing the limiting reactant, the theoretical yield (in grams), the percent yield, the remaining masses of any unreacted excess reagents, and whether the reaction is exothermic or endothermic along with its heat flow per mole.

Part 3: Data Abstraction

Representing Reality: Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track?

The core nouns our program needs to track are `ChemicalReaction`, `Reactant`, *and* `Product`.

Identifying Properties: For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?

Details I need: Chemical formula/name, molar mass, stoichiometric coefficient, and mass/moles.

Details I will ignore: Compound color, physical state appearance, container type, and manufacturer brand.

Part 4: Challenges, AI Strategy, & Goals

Difficulties: What parts of this project seem the most difficult or technically complex?

Parsing complex chemical formula strings to automatically compute molar masses and structuring the multi-step stoichiometric logic (comparing mole ratios and identifying the limiting reactant programmatically) will likely be the most challenging parts.

Coding Strategy: If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning"?

I will break the code down into smaller functions (e.g., mass-to-mole conversion, limiting reagent finder), trace each function line-by-line, write unit tests for edge cases, and manually verify the stoichiometry of test reactions using textbook examples before integrating them into the Drafter framework.

Goals: What is your ultimate goal for this project?

To build a fully functional, highly accurate chemistry calculator that I can actually use to verify my chemical engineering stoichiometry and thermodynamics homework.

Part 5: Research & Brainstorming Conversation

The Initial Prompt/Question: Paste your prompt below:

"I want to build a Python web tool using Drafter to help chemical engineering students calculate stoichiometry, limiting reactants, and reaction heat flow. What are 3 common computational challenges students face when writing stoichiometry scripts, and what are 3 core features a basic MVP tool should have?"

The Response: Briefly summarize the best ideas the AI or your friend suggested.

The AI suggested that key challenges include string-parsing of chemical formulas (handling subscripts and parentheses), floating-point precision errors during mole calculations, and handling reactions with multiple products. It recommended keeping the MVP focused on three core features: a clean input form for entering balanced equations and masses, a robust function to compute moles and identify the limiting reagent, and a clear output view displaying theoretical and percent yields.

The Pushback: Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. Paste your response here:

"Using advanced regex libraries or external APIs to automatically balance arbitrary chemical equations and pull thermodynamic data from online databases is too complicated for an introductory CS project because string parsing edge cases for nested parentheses will take too much time. What is a simpler alternative where the user inputs coefficients or predefined compounds?"

Reflection: Based on this conversation, how has your project idea changed or become more focused?

The conversation helped me narrow my scope so that instead of trying to build a fully automated equation balancer from scratch, I can focus on clean data structures (like dataclasses) and reliable stoichiometry logic using user-supplied coefficients and molar masses.

Part 6: Resource Planning

What resources will you use if you get stuck?

I will use course lecture notes, Python documentation, Drafter framework reference guides, and targeted LLM queries for debugging syntax errors.

If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

First, I will isolate the specific error message or stack trace. Second, I will check button bindings, route functions, and textbox parameter names in Drafter. Third, I will test the core calculation logic independently in a standard Python script before re-integrating it into the web interface.

Part 7: LLM Conversation

None

```
User: I want to build a Python web tool using Drafter to
help chemical engineering students calculate stoichiometry,
limiting reactants, and reaction heat flow. What are 3
common computational challenges students face when writing
stoichiometry scripts, and what are 3 core features a basic
MVP tool should have?

AI:
1. Common computational challenges:
   - Parsing chemical formulas with subscripts and
parentheses accurately into molecular weights.
   - Handling floating-point rounding inaccuracies during
mole conversions.
   - Structuring conditional logic to correctly compare
stoichiometric mole ratios for limiting reactant
```

identification.
2. Three core MVP features:
   - A clean input interface for entering reactant masses and stoichiometric coefficients.
   - A dedicated calculation module for mole conversion and limiting reagent identification.
   - An organized results display for theoretical yield, percent yield, and reaction enthalpy.

User: Using advanced regex libraries or external APIs to automatically balance arbitrary chemical equations and pull thermodynamic data from online databases is too complicated for an introductory CS project because string parsing edge cases for nested parentheses will take too much time. What is a simpler alternative where the user inputs coefficients or predefined compounds?

AI: A much simpler alternative is to create a dictionary or dataclass structure containing a curated list of common chemical compounds with preset molar masses, and have the user provide the balanced equation coefficients and mass inputs directly through textboxes, eliminating complex string parsing while keeping the focus on core stoichiometry and yield calculations.

[redacted]

Part 8: Submit

simpler altern

Gemini is AI and can make mistakes.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

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